# MN50753 Heuristics & Simulation
# Coursework II: Airport Baggage Handling System Simulation
# Individual Coursework Report

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Module: MN50753 Heuristics & Simulation
Word Count: 2,124 words

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# 1. Introduction and System Overview

## 1.1 Purpose of the Study

This study uses Arena simulation to evaluate the performance of an airport baggage handling system. The aim is to identify key bottlenecks and assess queue waiting times, queue lengths, resource utilisation, and total time in the system for standard baggage and baggage requiring additional security checks.

## 1.2 Model Overview

The model uses minutes as the base time unit. Each replication represents one week of operation, with seven 18-hour operating days. The simulation was run for 40 replications to capture variation in arrivals, service times and routing outcomes.

Each entity represents one passenger bag. After arrival, bags are assigned to a flight-zone group and then pass through security screening. Flagged bags are sent to manual inspection, where they are either cleared or removed from the system.

Cleared bags move to sorting. They are normally routed to their assigned zone, but may be temporarily redirected when congestion occurs. Their original flight-zone identity is still retained for later matching.

After sorting, bags enter reconciliation, where they are grouped by flight zone and released using batching and timeout logic. They then move to loading preparation and final scan. Bags that fail the final scan return to security screening with priority, while bags that pass continue to aircraft loading.

Aircraft loading is also based on flight-zone groups. After loading is completed, bags leave the system, and total time is recorded separately for standard bags and bags requiring additional security inspection. Staff breaks are represented through resource schedules and workload-based breaks.

# 2. Assumptions and Limitations

## 2.1 Key Assumptions

For congestion rerouting, bags are sent to the shortest sorting queue. If queue lengths are tied, a fixed priority rule is used: Zone A is preferred over Zone B when both are equally short, and Zone B is preferred over Zone C when both are equally short. Rerouting is also assumed to occur immediately without additional travel or handling time. Zone A, B and C were treated as flight groups rather than individual flight numbers.

Sorting staff take two 30-minute breaks during the 18-hour operating day. The first break was scheduled after five hours, as described in the case. A second break was

added later to avoid assuming that staff work almost continuously for the rest of the day. A similar rule was applied to security officers, but they take their breaks at the same time after six hours of work. Although the final working period of sorting staff is longer than five hours, this was considered more realistic than adding another break close to closing time, where staff would return and work only briefly before the system stops operating.

The model assumes that internal bag movements can occur immediately without additional travel time or handling delay. This applies to both congestion-based rerouting between sorting zones and bags that fail the final scan and return to security screening. In reality, these movements may depend on conveyor layout, staff coordination, physical transfer distance, and system re-registration. Therefore, the model may underestimate the time and operational effort required for rerouted or returned bags.

### 2.2 Limitations

Several limitations should be noted. First, machine breakdowns were not included in the model. Scanners, sorting equipment, and loading-related resources were assumed to be available whenever the relevant staff resources were available. In reality, equipment failures could temporarily reduce system capacity and increase queue lengths and waiting times. As a result, the model may underestimate delays during disrupted operating periods.

Second, staff availability was simplified by treating the 18-hour operating day as a single staffing period with scheduled breaks. In practice, it is unlikely that the same staff members would work the full operating day without shift rotation. A more realistic model would include multiple shifts, staff handover periods, and possibly different staffing levels across the day. This simplification may affect estimated resource utilisation and queue performance.

Third, bag arrivals were modelled using a constant interarrival distribution throughout the operating day. However, baggage arrivals are likely to vary according to flight schedules, with clear peak and off-peak periods. This means the model may underestimate queues during busy periods and overestimate congestion during quieter periods. This limitation is also linked to the simplified staffing assumption, because real airports may adjust staffing levels according to expected demand.

Finally, the simulation is based on a fixed operating period. At the end of the run, around 50 out of 150 bags remain in the system. Therefore, the average total time and waiting time may be underestimated, as unfinished bags are not fully included in the final output. The results should therefore be interpreted as system performance within a fixed operating period rather than the complete processing time for all bags.

### 2.3 Possible Model Improvements

Future model improvements could address these limitations. Machine breakdowns could be added using failure and repair logic for scanners, sorting equipment and loading-related resources. Staff availability could be improved by modelling multiple shifts, handover periods and different staffing levels across the day using more detailed resource schedules. Bag arrivals could also be made time-dependent, with higher arrival rates during flight peaks and lower rates during quiet periods. In addition, rerouted and returned bags could be given extra movement delays using Delay modules to represent conveyor switching, physical transfer and staff handling.

## 3. Base Model Results

### 3.1 Queue Waiting Times

| Performance measure | Mean (mins) | Mean of Maximum (mins) | 95% confidence interval | Key result |
|---|---|---|---|---|
| Zone A | 2.84 | 11.77 | [2.36, 3.32] | Low |
| Zone B | 3.33 | 11.72 | [2.49, 4.18] | Low |
| Zone C | 19.59 | 43.9 | [16.36, 22.82] | Bottleneck |
| Reconciliation A | 11.35 | 20 | [11.12, 11.57] | Reasonable |
| Reconciliation B | 11.69 | 20 | [11.49, 11.89] | Reasonable |
| Reconciliation C | 11.87 | 20 | [11.70, 12.04] | Reasonable |
| Loading Batch A | 19.92 | 52.34 | [18.49, 21.34] | Moderate |
| Loading Batch B | 23.14 | 55.97 | [21.14, 25.14] | Moderate |
| Loading Batch C | 44.13 | 85.6 | [41.82, 46.45] | Longest loading batch delay |
| Security Screening (New bags) | 0.47 | 3.85 | [0.38, 0.57] | Low |
| Security Screening (Return bags) | 0.27 | 0.77 | [0.18, 0.37] | Low |
| Manual Inspection | 0.39 | 3.09 | [0.25, 0.54] | Low |
| Loading Preparation | 1.91 | 8.32 | [1.73, 2.08] | Low |

*Table 1. Queue Waiting Time Summary*

The holding queues (A, B, and C) exhibit moderate waiting times of around 11–12 minutes. These values are below the 20-minute timeout limit, suggesting that the reconciliation batching logic is functioning reasonably and bags are not waiting excessively at this stage.

The loading batch queues also show relatively high average waiting times of around 20–44 minutes, suggesting that bags spend considerable time waiting to form batches before loading. The mean maximum waiting times are even higher, especially for Loading Batch C, which reaches around 85.6 minutes. This indicates that the loading batch stage can create severe delays in some replications. In contrast, the waiting times at the security screening and manual inspection stages remain low, indicating sufficient processing capacity at these stages.

Overall, the main bottlenecks are identified at Zone C and the loading batch stages, which contribute significantly to delays in the system.

### 3.2 Queue Lengths

| Performance measure | Mean (bags) | Mean of Maximum (bags) | 95% confidence interval | Key result |
|---|---|---|---|---|
| Zone A | 0.69 | 4.4 | [0.55, 0.83] | Low |
| Zone B | 0.75 | 3.9 | [0.52, 0.98] | Low |
| Zone C | 3.04 | 7.4 | [2.42, 3.66] | Accumulation |
| Loading Batch A | 3.46 | 10 | [3.28, 3.64] | Lower than required batch size |
| Loading Batch B | 3.48 | 10 | [3.31, 3.65] | Lower than required batch size |
| Loading Batch C | 3.28 | 10 | [3.12, 3.43] | Lower than required batch size |
| Reconciliation Hold A | 2.42 | 7.5 | [2.32, 2.52] | Low |
| Reconciliation Hold B | 2.16 | 6.5 | [2.06, 2.26] | Low |
| Reconciliation Hold C | 1.18 | 3 | [1.14, 1.21] | Low |

*Table 2. Queue Length Summary*

Queue length results show where bags physically accumulate in the system. A high average queue length indicates persistent congestion, while a high maximum queue length suggests occasional peak-period pressure.

Zone C has the highest average and mean of maximum queue length, indicating that bags accumulate more seriously at this stage. In contrast, Zone A and Zone B have much lower queue lengths because each zone has two workers, which appears sufficient to handle their workload. This suggests that Zone C’s single-worker

capacity is not enough, and adding one more worker to Zone C could help reduce congestion and improve overall sorting efficiency.

Consistent with the queue waiting time results, the loading batch queues show relatively low average accumulation, generally below the required batch size of 10. This indicates that bags spend considerable time waiting for enough bags to accumulate before the loading batch can be released. The maximum queue length reaches 10, which is reasonable because loading is triggered once the required batch size is reached.

The holding queue lengths remain relatively low and do not approach the full batch size of 15. This suggests that batches are unlikely to be released mainly by reaching the full batch size. Instead, the 20-minute timeout rule is likely to play an important role in releasing reconciliation batches. However, this should be interpreted as supporting evidence, because queue length measures the number of bags waiting at a point in time rather than the exact batch size at release.

### 3.3 Resource Utilisation

| Resource | Mean (%) | 95% confidence interval (%) | Key result |
|---|---|---|---|
| Sorting Staff A1 | 68.1 | [65.0, 71.1] | Reasonable |
| Sorting Staff A2 | 67 | [64.3, 69.7] | Reasonable |
| Sorting Staff B1 | 69.4 | [66.1, 72.8] | Reasonable |
| Sorting Staff B2 | 67.5 | [64.3, 70.7] | Reasonable |
| Sorting Staff C | 90.1 | [87.6, 92.5] | Overloaded |
| Officer 1 | 30.6 | [28.0, 33.2] | Underutilised |
| Officer 2 | 28.5 | [25.8, 31.2] | Underutilised |
| Loading Handler | 45.1 | [44.0, 46.3] | Reasonable |
| Loading Team | 24.3 | [23.4, 25.1] | Underutilised |

*Table 3. Resource Utilisation Summary*

Among the sorting staff in Zones A, B, and C, Zone A and Zone B show relatively balanced utilisation, with staff utilisation ranging from around 67% to 69%. However, Staff C has much higher utilisation at 90.1%, indicating that Zone C is overloaded compared with the other sorting zones. This imbalance suggests that Zone C has limited spare capacity and should be considered for improvement.

In the manual inspection area, the utilisation of each officer is relatively low, at around 29%–31%. This suggests that manual inspection has spare capacity, and the stage may still be able to operate with one officer if one officer is reassigned to support another bottleneck area.

The loading batch queues have relatively high waiting times, but the loading team utilisation is only 24.3%. This suggests that the delay before loading is mainly

caused by bags waiting to form a sufficient batch, rather than by a lack of loading team capacity.

The loading preparation handlers show moderate utilisation at around 45.1%. With three handlers scheduled, this suggests that the loading preparation area has sufficient capacity and is not a major bottleneck.

### 3.4 Average Total Time in System

| Baggage type | Mean total time (mins) | 95% confidence interval (mins) | Key result |
|---|---|---|---|
| Standard baggage | 73.51 | [72.27, 74.74] | Baseline total time |
| Additional-security baggage | 81.34 | [77.36, 85.32] | Longer due to manual inspection |

*Table 4. Total Time in System Summary*

The average total time in the system is 73.51 minutes for standard baggage and 81.34 minutes for baggage requiring additional security checks. This indicates that additional security baggage spends approximately 7.83 minutes longer in the system. This is expected because these bags require manual inspection before returning to the main baggage flow. However, the difference is not extremely large, suggesting that manual inspection is not the main source of delay.

## 4. Interpretation of Results

Overall, the main bottlenecks in the system are identified at Zone C and the loading batch stages. These areas contribute most significantly to delays in the system.

Based on the analysis, Zone C has the highest waiting time, the longest queue length, and the highest staff utilisation. This indicates that Zone C is the main sorting bottleneck. Since the manual inspection officers have relatively low utilisation, one officer could be reassigned to support Zone C to reduce queue waiting time and improve flow balance.

The loading teams are not fully utilised, even though the loading batch queues show noticeable waiting times. This suggests that the delay is mainly caused by bags waiting to meet the required batch size before loading can begin, rather than by insufficient loading team capacity. Therefore, adjusting the loading batch release rule may help improve workflow balance.

In contrast, the other stages, such as security screening, manual inspection, and loading preparation, show reasonable waiting times, queue lengths, and resource utilisation. Therefore, improvements should focus on reducing the workload at Zone

C and optimising the loading batch release policy. These adjustments are expected to improve the overall efficiency of the baggage handling system.

## 5. Alternative Configurations

### 5.1 Alternative 1: Reassign one manual inspection officer to Zone C

Reallocating labour makes better use of existing resources without adding staff. The small gap in total time between standard baggage and baggage requiring additional security checks shows that manual inspection has spare capacity, while Zone C is heavily loaded. Adding a worker to Zone C should therefore reduce congestion. Assuming that an officer can perform sorting tasks without additional training, the recommended change is to transfer one officer from manual inspection to Zone C sorting.

### 5.2 Alternative 2: Adjust the loading batch release rule

In order to improve the efficiency of the loading process, the main adjustment is to add a timer of around 15 or 20 minutes to reduce the average waiting time for releasing loading batches. This timer is proposed based on the observed average loading batch waiting time of around 20–40 minutes. Therefore, loading batches can be released earlier when the full batch size has not yet been reached. This is expected to increase the utilisation of the loading teams and make the loading process more efficient.

## 6. Implemented Alternative and Comparison

Reassigning one manual inspection officer to Zone C significantly reduces the Zone C waiting time, from around 19.59 minutes to 1.66 minutes. This confirms that additional capacity at Zone C is effective in reducing the main sorting bottleneck. The Zone C queue length also decreases substantially under the improved scenario, indicating a much smoother flow through this stage. Although the manual inspection waiting time increases from 0.39 minutes to 4.47 minutes due to the reduced number of officers, this increase remains relatively moderate. The average total time for standard baggage also decreases, while the total time for additional security baggage changes only slightly. Therefore, the reassignment appears to improve the overall system performance without creating a severe new bottleneck at manual inspection.

| Performance measure | Base model (mins) | Alternative model (mins) | Change (mins) | Key result |
|---|---|---|---|---|
| Zone C waiting time | 19.59 | 1.66 | -17.93 | Major improvement |
| Manual Inspection waiting time | 0.39 | 4.47 | +4.08 | Increases but remains manageable |
| Standard baggage total time | 73.51 | 70.72 | -2.79 | Improved |
| Additional-security baggage total time | 81.34 | 81.72 | +0.38 | Almost unchanged |
| Loading Batch C waiting time | 44.13 | 34.26 | -9.87 | Downstream delay reduced |

*Table 5. Comparison of Base Model and Reassigned Officer Alternative*

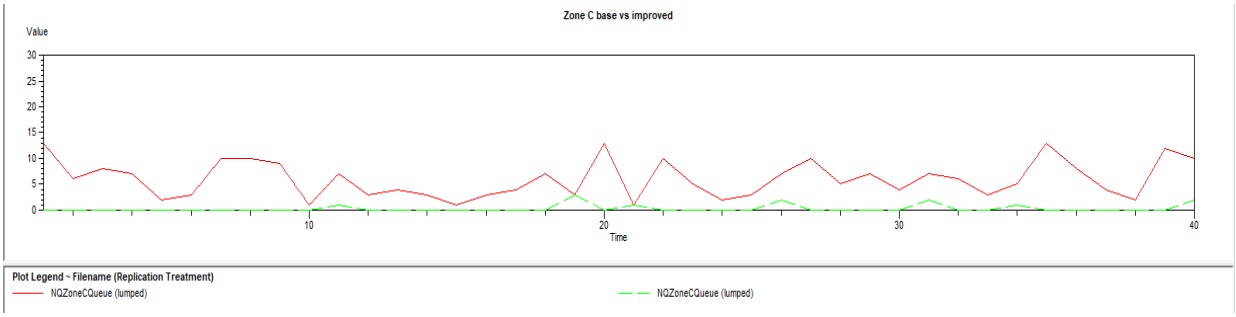



*Figure 1. Queue length of Zone C: base (red) vs improved (green)*

## 7. Conclusion

Overall, this study shows that simulation is a useful method for identifying bottlenecks in a complex system. By analysing waiting times, queue lengths, resource utilisation, and total time, the model helps locate where delays occur more accurately. It also allows improvement options to be tested before making real operational changes, reducing the cost and risk of trial and error. Although the model includes assumptions and cannot fully represent the flexibility of real airport operations, it still provides valuable evidence to support data-driven process improvement.